

The New Methods I

Crystal Structure • Density • Vacancy • Diffraction

VSEPR-SIM thenewmethods

Notation that is readable, compact, invertible where possible, and grouped as systems where that helps.

Crystal Structure Relations

FCC

$$\text{FCC:} \quad \left\{ \begin{array}{l} n = 4 \\ \text{CN} = 12 \\ \text{APF} = 0.74 \\ a = 2\sqrt{2} R \end{array} \right. \quad \Longleftrightarrow \quad R = \frac{\sqrt{2}}{4} a$$

BCC

$$\text{BCC:} \quad \left\{ \begin{array}{l} n = 2 \\ \text{CN} = 8 \\ \text{APF} = 0.68 \\ a = \frac{4R}{\sqrt{3}} \end{array} \right. \quad \Longleftrightarrow \quad R = \frac{\sqrt{3}}{4} a$$

HCP (for review, not default assumption)

$$\text{HCP:} \quad \left\{ \begin{array}{l} n = 6 \\ \text{CN} = 12 \\ \text{APF} = 0.74 \\ a = 2R \end{array} \right. \quad \text{ideal ratio:} \quad \frac{c}{a} \approx 1.633$$

Density Relations

Master equation for cubic unit cells:

$$\rho = \frac{nA}{N_A a^3}$$

where n = atoms per unit cell, A = atomic weight, N_A = Avogadro's number, a^3 = unit cell volume.

Inverse Forms

Solve for atomic weight:

$$A = \frac{\rho N_A a^3}{n}$$

Solve for lattice parameter:

$$a = \left(\frac{nA}{\rho N_A} \right)^{1/3}$$

One line, invertible, no theatrical suffering.

Vacancy Relations

Forward Form

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$$

Vacancy Fraction (often the more useful version)

$$\frac{N_v}{N} = \exp\left(-\frac{Q_v}{kT}\right)$$

Inverse Form for Q_v

Take the natural log:

$$\ln\left(\frac{N_v}{N}\right) = -\frac{Q_v}{kT} \quad \Rightarrow \quad Q_v = -kT \ln\left(\frac{N_v}{N}\right)$$

Inverse Form for Temperature

$$T = -\frac{Q_v}{k \ln(N_v/N)}$$

Memorization form: *exponential forward, log backward.*

Diffraction Relations

Bragg Law

$$n\lambda = 2d \sin \theta$$

Inverse Forms

Solve for spacing:

$$d = \frac{n\lambda}{2 \sin \theta}$$

Solve for angle:

$$\theta = \sin^{-1} \left(\frac{n\lambda}{2d} \right)$$

When XRD reports 2θ directly:

$$\theta = \frac{1}{2}(2\theta)$$

Cubic Plane-Spacing Relations

For cubic crystals:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad \Longleftrightarrow \quad a = d_{hkl} \sqrt{h^2 + k^2 + l^2}$$

Combined Diffraction System

Useful when solving structure problems:

$$\begin{cases} n\lambda = 2 d_{hkl} \sin \theta \\ d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \end{cases}$$

Substitute and solve for a :

$$a = \frac{n\lambda \sqrt{h^2 + k^2 + l^2}}{2 \sin \theta}$$

That is the one-line usable relation to keep.

Ionic Characterization

Let $\Delta\chi = |\chi_A - \chi_B|$.

$$\%IC = 100(1 - \exp[-\frac{1}{4}(\Delta\chi)^2])$$

System form (memorization form):

$$\begin{cases} \Delta\chi = |\chi_A - \chi_B| \\ \%IC = 100\left(1 - \exp\left[-\frac{(\Delta\chi)^2}{4}\right]\right) \end{cases}$$

Structure Classification Logic

Input: $\{R, A, \rho, T, \theta, \lambda\}$. Try FCC and BCC; for each compute $a(R)$, $\rho_{\text{theoretical}}$, d_{hkl} , and consistency with T .

Best structure via weighted residual:

$$S^* = \arg \min_{S \in \{\text{FCC}, \text{BCC}\}} [w_\rho |\rho_{\text{exp}} - \rho_S| + w_d |d_{\text{exp}} - d_S| + w_T \Pi_T(S)]$$

General classification objective:

$$J(S) = \sum_i w_i |x_i^{\text{measured}} - x_i(S)| \quad S^* = \arg \min_S J(S)$$

Measured features: $x_i \in \{\rho, d_{hkl}, a, \text{CN}, \text{APF}, T\text{-stability}\}$.

Ring Orbitals (compact Hückel form)

For a cyclic π -system with N p-orbitals:

$$E_m = \alpha + 2\beta \cos\left(\frac{2\pi m}{N}\right), \quad m = 0, 1, \dots, N-1$$

Fill from lowest E_m upward, 2 electrons per orbital.

$$4n + 2 \text{ pi electrons} \Rightarrow \text{aromatic} \quad 4n \text{ pi electrons} \Rightarrow \text{antiaromatic tendency}$$

Compact Memorization Block

$$\text{FCC: } a = 2\sqrt{2} R, \quad n = 4, \quad \text{CN} = 12, \quad \text{APF} = 0.74$$

$$\text{BCC: } a = \frac{4R}{\sqrt{3}}, \quad n = 2, \quad \text{CN} = 8, \quad \text{APF} = 0.68$$

$$\text{HCP: } a = 2R, \quad n = 6, \quad \text{CN} = 12, \quad \text{APF} = 0.74, \quad c/a \approx 1.633$$

$$\rho = \frac{nA}{N_A a^3}$$

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right) \quad Q_v = -kT \ln\left(\frac{N_v}{N}\right)$$

$$n\lambda = 2d \sin \theta \quad d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$\Delta\chi = |\chi_A - \chi_B| \quad \% \text{IC} = 100\left(1 - e^{-(\Delta\chi)^2/4}\right)$$

$$S^* = \arg \min_S J(S)$$

Full System Form

$$\left\{ \begin{array}{l} \text{Given: } A, R, \rho_{\text{exp}}, \lambda, \theta, T, \chi_A, \chi_B \\ a_{\text{FCC}} = 2\sqrt{2} R, \quad a_{\text{BCC}} = \frac{4R}{\sqrt{3}} \\ \rho_{\text{FCC}} = \frac{4A}{N_A a_{\text{FCC}}^3}, \quad \rho_{\text{BCC}} = \frac{2A}{N_A a_{\text{BCC}}^3} \\ d = \frac{n\lambda}{2 \sin \theta} \\ \Delta\chi = |\chi_A - \chi_B| \\ \% \text{IC} = 100\left(1 - e^{-(\Delta\chi)^2/4}\right) \\ \frac{N_v}{N} = \exp\left(-\frac{Q_v}{kT}\right) \\ S^* = \arg \min_{S \in \{\text{FCC}, \text{BCC}\}} J(S) \end{array} \right.$$